Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

Team Name	FAIX
Project / Solution Name	THE SMART-DEFECT TRACKER
Name (Leader)	CHE AHMAD RAHIMI BIN CHE ROZAID
Email & Phone (Leader)	rahimirozaid99@gmail.com 011-2566-0584
University Affiliation(s)	UNIVERSITI TEKNIKAL MALAYSIA MELAKA (UTeM)
Live Prototype URL	https://orbyte-defect-tracker-team-alpha.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/rahimisss2/orbyte-defect-tracker-team-alpha

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Our system runs on a streamlined Python 3.12 architecture. The front end is built with Streamlit giving staff and managers an intuitive interface to log defects, check the control room and sign off on claims.

The main backend logic (src/main.py) coordinates everything. When an incident is logged, it calls the Google Gemini 2.5 Flash API using secure Replit Secrets to analyze the suppliers history and draft a claim letter. Because API latency can happen we built a local deterministic backup engine as a fallback. This guarantees the application stays up even if the internet drops.

For data storage, we kept it lightweight by using a local synthetic_db.json file. To prevent multi-user bugs, we implemented a custom file-locking layer that uses atomic replacement to stop write conflicts. The entire system is hosted on Replit, which serves the app publicly as a live prototype.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.12	Core programming language used to build the entire backend engine, data routine and workflow scripts	Python Software Foundation (PSF) License
Streamlit Framework	Frontend UI framework used to construct the interactive screens, dashboards and manager interfaces	Apache License 2.0

Name & Version	Purpose in Project	License
Google Gemini API	AI decision model layer used to evaluate inident logs, process metrics and draft contextual claim letters	Google API Terms of Service

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API Provider: Google	Powers the smart decision layer to evaluate supply chain discrepancies, analyze incident metrics and generate contextual draft text for supplier claim letters	Free Tier
Replit Cloud Hosting Provider: Replit	Serves as the cloud development IDE and active hosting infrastructure, utilizing Asia servers to host the public web link.	Sponsor-provided

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit Deployments Provider: Replit	Main cloud hosting platform used to containerize the code and host the live Streamlit web application so users can access the interface online	Autoscale Plan / Asia Server Region
Replit Environment Secret Provider: Replit	Secure key management system used to inject the GEMINI_API_KEY straight to the apps working memory keeping the sensitive key completely hidden from public version control history	Built in Free Tier Feature / Global Region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	**Source / Provider**	**Usage (pre-trained / fine-tuned / self-trained)**	**License / Terms**
Gemini 2.5 Flash	Google	Pre Trained (Called via API to analyze defect logs and draft contextual claims)	Google API Terms of Service / Developer Terms

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	**Source / URL**	**Type (public / private / self-collected / synthetic)**	**License**	**Personal Data? (Y/N)**
Operational Defect DB	Local project repository Asset	Synthetic	Apache License 2.0	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit AI Agent	Assisting with live workspace environment setups, organizing core applications routines, structuring codebase scaffolding and fixing cloud configuration and deployment script errors dynamically	Major
ChatGPT	Structuring documentation outlines, proofreading technical project text and organizing submission records	Moderate
Gemini	Running initial design mockups, brainstorming error handling concepts for code latency, and drafting early sample templates for our documentation.	Minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

We treated AI like an assistant, meaning we human verified everything it touched. For the technical scaffolding and cloud script fixes we ran hands on tests in our staging environment to ensure the code was stable and secure. For the written side like documentation outlines and proofread text we carefully edited the content to ensure it fit our specific project context. We took total responsibility for the final delivery by cross checking all submission records manually ensuring no unverified AI content made it into our final submission.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☐ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Cloud Development IDE & Deployment Platform	Provided the complete sandboxed cloud workspace used by our team for collaborative code editing, local file management and instant live application hosting through a secure public link.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was important to the development of our prototype because it provided an online coding environment where our team could work together, manage project files, run and test the application. Its built in deployment feature also allowed us to make the prototype accessible through a public link. However, Replit was not essential to the core functionality of the application itself as the project could be developed and hosted using alternative platforms or local development tools. Using Replit mainly made the development process more convenient, efficient and collaborative for our team.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

The application collects and processes non-sensitive operational inventory data, specifically supplier names (e.g., Pak Ali Eggs Enterprise), defect types, and financial loss amounts. This structured information is temporarily sent to a third-party service, the Google Gemini API, solely to generate the contextual claim text paragraph.

No personal user data or sensitive human credentials are collected, stored, or transmitted. To ensure complete privacy and security compliance, the Google Gemini API key is securely contained inside Replit Secrets and loaded dynamically in-memory, preventing any hardcoded exposures in the source repository. All operational transaction logs are stored locally within a secure JSON file database. Data processed during runtime is transient and is not retained by the third-party AI provider for model training.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

The application may have some limitations such as providing inaccurate or incomplete results depending on the quality of the information entered and the systems performance. The AI featured may sometimes be incorrect or biased so users should review the information before making important decisions. The application may also experience delays or temporary downtime due to technical issues or service availability. To reduce these risks, the system is designed to provide clear and consistent results and users are encouraged to verify important information. Regular testing and maintenance can help identify errors and improve the application's reliability. The application also depends on the availability of its supporting services so some features may not work properly if those services are unavailable.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

√ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.

√ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.

√ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.

√ Any use of generative-AI tools for development has been disclosed and reviewed by the team.

√ Any use of official SDC sponsor resources has been declared in Section 5.

√ The prototype described here is accessible to judges through the links provided in Section 1.

√ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

Rahimi

[Date]
24/10/2026